



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 1.8.2026

In addition to Part I (General Handout for all courses, appended to the Timetable), this part provides further specific details regarding the course.

Course Number	CE F312
Course Title	Hydraulic Engineering
Instructor-in-charge Lecture L1 M W3 Th9 1232	Prof. SHIBANI KHANRA JHA Email: shibani@pilani.bits-pilani.ac.in Chamber No.- 1210H FDI
[Team of Instructors]	Rahul (RS), Ankit Pandey (RS), Utkarsh Chaudhry (RS), Arjun Gautam (RS), Aswathy Sathyan (RS), N Archana (RS)
Tutorial M1 1220, 1222	Rahul (RS), Ankit Pandey (RS)
Practical 1 Instructors Th 7 8 1114	Utkarsh Chaudhry (RS), Arjun Gautam (RS), Aswathy Sathyan (RS), N Archana (RS)]
Practical 2 Instructors F 1 2 1114	Rahul (RS), Arjun Gautam (RS), Ankit Pandey (RS), Aswathy Sathyan (RS)

1. Course Description:

This course introduces Civil Engineering students to the core concepts of hydraulic engineering, focusing on the **Behaviour of real fluids: boundary layer theory, turbulent flow through conduits; analysis of closed-conduit hydraulic systems including pipes, valves, fittings, and pumps, water hammer in pipes, pipe networks analysis: Hardy cross method; Open channel hydraulics: uniform and non-uniform flow; flow past immersed bodies: drag and lift; Analysis of Impact of jets; Introduction to fluid machinery.**

2. Scope & Course Objective:

The problems encountered by our society, such as water supply, irrigation, navigation, and water power, resulted in the development of Fluid Mechanics and Hydraulic Engineering. The objective of this course is to equip students with a fundamental understanding of the principles of fluid mechanics as applied to hydraulic systems and structures. The course aims to achieve the following **Course Objectives** (Teacher's perspective), aligned with the **Course Outcomes (COs)** (Students' perspective):

1. To develop an analytical understanding of flow behavior in pipes and channels under varying conditions.
2. To apply fluid mechanics principles in analyzing hydraulic systems, including energy losses and force impacts.
3. To introduce the working principles and performance of fluid machinery relevant to civil engineering.
4. To prepare students for practical problem-solving in the analysis, design, and assessment of water infrastructures of Pipe networks, open channels, and structures submerged in fluid, like bridge piers, dealt with by civil engineers.

3. Course Learning Outcomes (CLOs):

Upon the completion of this course, students will be able to perform the following:



CLO1. Explain the behaviour of real fluids, including boundary layer development and turbulent flow through conduits, and apply these concepts to practical hydraulic engineering problems.

CLO2. Analyze closed-conduit hydraulic systems by evaluating head losses, pipe flow characteristics, pumps, valves, fittings, and the effects of water hammer in pressurized pipelines.

CLO3. Analyze and solve pipe network problems using the Hardy Cross Method for the design and operation of water distribution systems.

CLO4. Analyze uniform and non-uniform flow in open channels using fundamental hydraulic principles and empirical equations for engineering design and performance assessment.

CLO5. Evaluate fluid forces on immersed bodies, including drag, lift, and the impact of jets on stationary and moving surfaces, for engineering applications.

CLO6. Explain the classification, operating principles, and applications of fluid machinery, particularly hydraulic turbines and pumps, and relate them to hydraulic engineering systems.

4. Textbooks and Reference Books:

- T1. Modi, P.N. and Seth, S.M., Hydraulics and Fluid Mechanics, Standard Book House, 19th ed., 2013.
 T2. Fox, R.W. and McDonald, A.T., Introduction to Fluid Mechanics, John Wiley and Sons Inc., Singapore, Eighth Edition, 2012.
 T3. Moondra, H.S., Gupta, R., Lab. Manual for Civil Engineering, CBS publishers & Dist, 2nd ed., 2000.
 R1. Douglas J. F., Gasiorek J. M., Swaffield J. A., and Jack L.B., Fluid Mechanics, Pearson Education, Second Impression, 2009.
 R2. Munson B.R., D.F. Young, T.H. Okiishi, Fundamentals of Fluid Mechanics, John Wiley and Sons (Asia) Pte. Ltd., Sixth Edition, 2009.
 R3. Chanamala Ratnam and Arun Vikram Kothapalli, Fluid Mechanics and Machinery, I K International Publishing House, New Delhi, 2010
 R4. Patra, K.C., Hydrology and Water Resources Engineering, Narosa Publishing House, 2001.

5. Course Format: Weekly Plan (for 14 Weeks)

Week No.	Mod No.	Description – Lecture Sessions	Reference (21st Ed., 2017)	Learning Outcomes
1	I	L1: Introduction to Hydraulic Engineering; Behaviour of real fluids. L2: Laminar and turbulent flow; Reynolds number. L3: Pipe flow equations and friction losses. Major and minor losses in pipes.	Chapter: Turbulent Flow in Pipes	Explain the behaviour of real fluids and distinguish between laminar and turbulent flows;
2	I	L4: Hydraulic Grade Line (HGL) and Total Energy Line (TEL). L5: Velocity profiles in turbulent flow. L6: Turbulent flow through conduits and engineering applications.	Chapter: Turbulent Flow in Pipes	Evaluate head losses and energy distribution in pressurized pipe systems. Analyze turbulent flow characteristics in conduits.
3	II	L7: Boundary layer theory, formation and thickness.	Chapter: Boundary Layer	Describe boundary layer development.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Week No.	Mod No.	Description – Lecture Sessions	Reference (21st Ed., 2017)	Learning Outcomes
		L8: Boundary layer separation and control. L9: Boundary layer in Pipes and external flow systems	theory	Apply boundary layer concepts
4	III	L10: Pipes in series and parallel. L11: Valves and fittings in hydraulic systems. L12: Analysis of closed-conduit hydraulic systems.	Chapter: Flow through pipes	Analyze closed-conduit hydraulic systems consisting of pipes, valves and fittings.
	III	L13: Pumps in hydraulic systems; system characteristics. L14: Operating point of pumps and pipelines. L15: Numerical problems on closed-conduit systems.	Chapter: Flow through pipes (Introduction to Fluid Machinery)	Analyze the performance of pumps integrated into hydraulic systems.
6	III	L16: Water hammer in pipes. L17: Pressure rise due to valve closure. L18: Control measures and surge protection.	Chapter: Flow through pipes (Water Hammer)	Evaluate water hammer effects and recommend Suitable control measures.
7	III	L19: Pipe network analysis—Introduction. L20: Hardy Cross Method—Procedure. L21: Numerical problems using Hardy Cross Method.	Chapter: Flow through pipes (Pipe Network)	Solve hydraulic pipe network problems using the Hardy Cross Method.
8	IV	L22: Open channel flow—Classification. L23: Uniform flow—Chezy and Manning equations. L24: Design examples on uniform flow.	Chapter: Open Channel Flow – Uniform & Non-uniform	Analyze uniform flow in open channels using empirical equations.
9	IV	L25: Specific energy and critical flow. L26: Non-uniform flow concepts. L27: Gradually varied flow and applications.	Chapter: Open Channel Flow – Uniform & Non-uniform	Explain non-uniform flow phenomena and determine flow conditions in open channels.
10	IV	L28: Hydraulic jump. L29: Energy dissipation and practical applications. L30: Numerical problems on open channel hydraulics.	Chapter: Open Channel Flow – Uniform & Non-uniform	Evaluate hydraulic jumps and their engineering applications.
11	V	L31: Flow past immersed bodies. L32: Drag force and drag coefficient. L33: Lift force and engineering	Chapter: Drag and Lift	Determine drag and lift forces acting on immersed bodies.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Week No.	Mod No.	Description – Lecture Sessions	Reference (21st Ed., 2017)	Learning Outcomes
		applications.		
12	VI	L34: Impact of jets on stationary plates. L35: Impact of jets on moving vanes. L36: Numerical problems and applications.	Chapter: Impact of Jet	Analyze forces developed due to the impact of jets on stationary and moving surfaces.
13	VII	L37: Introduction to fluid machinery and classification. L38: Overview of hydraulic turbines and pumps. L39: Course review and integrated problem solving.	Chapter: Introduction to Fluid Machinery	Classify fluid machinery and relate hydraulic principles to the operation of turbines and pumps.
14		Doubt Classes	-	-

6. Practical Details:

S No	Name of Experiments	Reference T3
1.	Darcy's Friction factor 'f' of pipes of different diameter pipes	3.5
2.	Discharge through an orifice under constant head	3.6
3.	Discharge through an orifice under varying head	3.7
4.	The coefficient of discharge 'C _d ' of the V notch and rectangular notch, and to plot the calibration curve	3.8
5.	Study of the formation of a hydraulic jump	3.10
6.	i) Verify Stokes' law ii) To study the type of flow using Reynolds's apparatus	3.2 Suppl. Notes
7.	i) To demonstrate two-dimensional laminar flow around differently shaped models using Hele-Shaw apparatus ii) Study of Metacentric Height Apparatus	Suppl. Notes
8.	Study of the Impact of Jet	Suppl. Notes
9.	Study of Specific Energy Curve using Multi-purpose Flume	Suppl. Notes
	Demonstration of water hammer problems	Suppl. Notes

Lab Component Remarks:

- Each Lab (40M): i) Hands-On (10M) ii) Report (10M) iii) End Sem Quiz (20M)
- Use a standard practical notebook for report writing.
- Each student must write the reading of each experiment on A4 paper and get it signed by the instructor and student on the day of the experiment itself. This signed page must be attached in the report notebook.
- The report must be submitted in the next class within the first 10 minutes of lab class.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

3. Penalty of -2M will be allotted for i) not attaching the signed reading, ii) not submitting within the first 10 minutes of lab class, iii) not using the proper graph (arithmetic or logarithmic) wherever applicable.

7. Evaluation Schemes:

Components	Duration (Minutes)	Weightage % (Marks)	Date & Time	Component
Mid-Semester	90 min	20 (40M)	07/10/2026 AN1 AN1: 2:00 to 3:30 P.M.	CB
Comprehensive Examination	180 min	30 (60M)	10/12/2026 FN FN: 8.00 to 11.00 A.M.	CB+OB
Class Assessment	Class hour	10 (20M)	TBA	CB/OB
Tutorial	Tutorial Hour	10 (20M)	TBA	CB/OB
Lab + Field Practical		30 (60M) Details i) Hands-On (10M) ii) Report (10M) iii) End Sem Quiz (20M) iv) Field Practical (20M)	As per the timetable	OB

Criterion for NC: [30% of the average of the top three performers or 40% of the median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.]

Remarks: No makeup is permitted for Class assessment, Tutorial, Lab and Practical classes

- 8. Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components. The 10% class assessment is linked to attendance.
- 9. Chamber Consultation Hour:** Consultation Hour – **Monday 11:00-12:00 am**. Note: IC can be reached through email or in person in the chamber.
- 10. Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for the Mid-sem or comprehensive exam only.

Prof. Shibani Khanra Jha
Instructor-in-Charge
CEF312